h-adic topology and profinite linear lambda terms

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1 Profinite Lambda terms and related concepts

One starts with the observation that if we have a work over a finite alphabet Σ:

$$W = a_{w_1} \dots a_{w_n} \in \Sigma^*$$

then one can represent W as a well-typed term in the simply typed lambda calculus:

$$a_{w_1} : q \to q, \dots, a_{w_n} : q \to q \vdash W : q \to q$$

where q is a base type that we informally interpret as a type of states, so we have a closed lambda term

$$\lambda a^{q\to q}_{w_1} \dots \lambda a^{q\to q}_{w_n} \lambda x^q . a_{w_k}(\dots(a_{w_1}\ q)\dots) : (q \to q) \dots (q \to q) \to q \to q$$

encoding a word w via the instance of the Church encoding. Further, we will call the type of such terms Church_Σ. Further, we have the monoid isomorphism:

$$\Sigma^* \cong \{M \text{ is a lambda term } | M : \text{Church}_\Sigma \}/\beta\eta$$

In other words, words over a finite alphabet can be viewed as lambda terms of a particular type factorised module the $\beta\eta$-equivalence. Generally, there is amount literature studying both syntactic and semantic aspects of the Church encoding of the finite words.

The claim made in [vGMM23] connects the finitary interpretation of STLC in the category of finite sets **FinSet** with deterministic finite automata as follows. Take any finite set Q, so for any type A, we define a set $[\![A]\!]_Q$ inductively defined as

- $[\![p]\!]_Q = Q$
- $[\![A \to B]\!]_Q = [\![A]\!]_Q \Rightarrow [\![B]\!]_Q$

so every term $x_1 : A_1, \ldots, x_n : A_n \vdash M : A$ transports to the function between finite sets $[\![M]\!] : [\![A_1]\!]_Q \times \ldots \times [\![A_n]\!]_Q \to [\![A]\!]_Q$. On closed terms, the interpretation induces a function $[\![.]\!]_Q : \Lambda_{\beta\eta}(A) \to [\![A]\!]_Q$ mapping every closed term M of type A to its interpretation $[\![M]\!]_Q$. Further, a word $W = a_{w_1} \ldots a_{w_k} \in \Sigma^*$ is mapped to the following morphism in **FinSet**: $[\![W]\!]_Q : (Q \Rightarrow Q) \times \ldots \times (Q \Rightarrow Q) \to (Q \Rightarrow Q)$, i.e.,

$$[\![W]\!]_Q : f_{w_1}, \ldots, f_{w_k} \mapsto f_{w_k} \circ \ldots \circ f_{w_1}$$

Recall that a *deterministic finite state automaton* on a finite alphabet $\Sigma = \{a_1, \ldots, a_n\}$ is a structure $\mathcal{A} = (Q, \delta, q_0, Acc)$, where Q is a finite set of states, $\delta : \Sigma \times Q \to Q$ is a transition function, $q_0 \in Q$ is the initial state and $Acc \subseteq Q$ is the set of accepting states. The transition function δ induces a family of functions $\delta_{a_i}(q_k) = \delta(a_i, q_k) : Q \to Q$ for each $a_i \in \Sigma$ and $q_k \in Q$. $\delta_{a_i}(q_k) = q_{k'}$ means that $\mathcal{A}$ transitions from q_k to q'_k encountering a_i. Now one can apply the interpretation $[\![W]\!]_Q$ to the functions $\delta_{a_{w_i}}$ for $1 \leq i \leq k$, so we obtain

$$\delta_w : [\![W]\!]_Q(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}}) : Q \to Q$$

Therefore, given a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, Acc)$, the tuple (Q, δ, q_0) induces the evaluation function

$$\mathrm{eval}_{\delta, q_0} : [\![\mathrm{Church}_\Sigma]\!]_Q \to Q$$

transporting every functional $F \in [\![\mathrm{Church}_\Sigma]\!]_Q$ to the state $F(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}})$ in Q, therefore we have

$$\Sigma^* \cong \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q \xrightarrow{\mathrm{eval}_{\delta, q_0}} Q$$

so we associate every finite word $w \in \Sigma^*$ with the final state $q_f = \delta_w(q_0)$. The inverse image $Acc \subseteq Q$ under this composite is called a *regular language* $L_\mathcal{A}$, the set of finite words recognisable by $\mathcal{A}$.

Regular languages $L_\mathcal{A}$ form a Boolean algebra $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ of regular languages of lambda terms of type Church_Σ recognised by Q. The algebra can be defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![\mathrm{Church}_\Sigma]\!]} \to 2^{\Sigma^*}$ obtained by applying the contravariant powerset functor $2^{(.)} : \mathbf{FinSet}^{op} \to \mathbf{BA}$ to the semantic bracket $[\![.]\!]_Q$. The Boolean algebra $\mathrm{Reg}_Q(A)$ of regular languages of lambda terms of any higher order type A recognisable by a finite set Q is defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![A]\!]_Q} \to 2^{\Lambda_{\beta\eta}(A)}$. That is, a set L of lambda terms of type A is recognisable by the finite set Q exactly when it has the form $[\![.]\!]_Q^{-1}(\mathrm{Acc}) = \{M \in \Lambda_{\beta\eta}(A) \,|\, [\![M]\!]_Q \in \mathrm{Acc}\}$ for some choice $\mathrm{Acc} \subseteq [\![A]\!]_Q$ of accepting elements. Thus, the collection $\mathrm{Reg}(A) \subseteq 2^{\Lambda_{\beta\eta}(A)}$ of lambda terms of type A as follows

$$\mathrm{Reg}(A) = \bigcup \{\mathrm{Reg}_Q(A) \,|\, Q \in \mathbf{FinSet}\}.$$

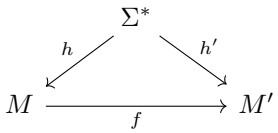
The monoid $\widehat{\Sigma^*}$ is the free profinite monoid generated by Σ. $\widehat{\Sigma^*}$ can be constructed as the limit, computed in the category of monoids **Mon**, of the projective system of finite monoid homomorphisms

$$(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi \subseteq \phi'}$$

where ϕ and ϕ' are finite index congruences on the free monoid Σ^* such that $\phi \subseteq \phi'$. Every finite index congruence ϕ, in turn, induces a surjective homomorphism $h : \Sigma^* \to M$ for the quotient monoid $M = \Sigma^*/\phi$. The surjectivity condition on h can be relaxed to show that $\widehat{\Sigma^*}$ of profinite words is the codirected limit of the composite functor

$$\Sigma^*/\mathbf{FinMon} \xrightarrow{\pi} \mathbf{FinMon} \hookrightarrow \mathbf{Mon}$$

$\Sigma^*/\mathbf{FinMon}$ is the slice category whose objects (M, h) are pairs consisting of a finite monoid M and a homomorphism $h : \Sigma^* \to M$ and whose morphisms $(M, h) \to (M', h')$ are homomorphisms $f : M \to M'$ such that the following triangle commutes:



The projection functor $\pi : \Sigma^*/\mathbf{FinMon} \to \mathbf{FinMon}$ transports (M, h) to its underlying monoid M. One obtains $\widehat{\Sigma^*}$ as the limit of a codirected diagram of monoid homomorphisms $(M \to M')_{(M,h)\to(M',h')}$ extending the diagram formed from the family $(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi\subseteq\phi'}$ to all homomorphisms $h : \Sigma^* \to M$ to a finite monoid M.

Recall that every homomorphism $h : \Sigma^* \to M$ induces a deterministic finite automaton with $Q := M$, $\delta(a, q) = q \cdot_M a$ for every $q \in M$ and $a \in \Sigma$. So every homomorphism $h : \Sigma^* \to M$ (assume $M = \{q_1, \ldots, q_m\}$) induces a decomposition of Σ^* into m components $L_{q_i} = h^{-1}(q_i)$ for $1 \leq i \leq m$ where each L_{q_i} is a regular language. Let $\mathrm{Reg}_{(M,h)}(\Sigma)$ denote the Boolean algebra of languages of the form $L_q = h^{-1}(q)$ for $q \in M$. We have a functor:

$$\mathrm{Reg}_{(.)}(\Sigma) : (\Sigma^*/\mathbf{FinMon})^{op} \to \mathbf{BA}$$

The Boolean algebra $\mathrm{Reg}_{(M,h)}(\Sigma)$ coincides with 2^h, the image of $h : \Sigma^* \to M$ applied to the contravariant powerset functor. Also, the following directed diagram in $\mathbf{BA}$ associated to $\mathrm{Reg}_{(.)}(\Sigma)$:

$$(\mathrm{Reg}_{(M',h')}(\Sigma) \to \mathrm{Reg}_{(M,h)}(\Sigma))_{(M,h)\to(M',h')}$$

can be obtained by applying the contravariant powerset functor to the above trianlge and $(M \to M')_{(M,h)\to(M',h')}$.

1.1 The Explicit Description of Profinite Topology

This description is based on [Pin10, C.X.2]. Let Σ be a finite alphabet and let M be a finite monoid, given $u, v \in \Sigma^*$, a monoid homomorphism $f : \Sigma^* \to M$ *separates* u and v if $f(u) \neq f(v)$. A monoid M separates u and v if there is a homomorphism $\Sigma^* \to M$ separating u and v. The important fact is that every u and v are separated by some monoid M.

A metric on Σ^* is defined as follows: two words are closed if they are separated by the large monoid. Take $u, v \in \Sigma^*$, put:

- $r(u, v) = \min\{|M| \mid M \text{ separates } u \text{ and } v\}$,
- $d(u, v) = 2^{-r(u,v)}$.

Therefore, $d(u, v) < 2^n$ iff u and v cannot be separated by a monoid smaller than n and $d(u, v) < \varepsilon$ iff u and v cannot be separated by a monoid of size n, where n is the unique natural number such that $2^{-(n+1)} < \varepsilon \leq 2^{-n}$. d is a metric on Σ^* such that (Σ^*, d) is an ultrametric space such that, for each $u, v, w \in \Sigma^*$:

- $d(uw, vw) \leq d(uv)$,
- $d(wu, wv) \leq d(uv)$.

As a topological space, (Σ^*, d) is discrete, but open balls are of more interest. For any $n \geq 2$, consider the congruence relation $\sim_n$ on Σ^* defined as:

$$u \sim_n v \Leftrightarrow \forall M\ (|M| \leq n \to (\forall f : \Sigma^* \twoheadrightarrow M) \to f(u) = f(v))$$

Clearly $\sim_n$ is a congruence of finite index as they are finitely many surjections $\Sigma^* \twoheadrightarrow M$ for a fixed finite M. Thus $d(u, v) < 2^{-n}$ iff they cannot be separated by a monoid of size smaller than n.

Proposition 1.1. Let Σ be a finite alphabet,

1. The concatenation function $(u, v) \mapsto uv$ is uniformly continuous,
2. Every open ball of (Σ^*, d) is a recognisable language and every recognisable language is a finite union of open balls.
3. (Σ^*, d) is totally bounded: for every $n > 0$, Σ^* is covered by a finite number of open balls of radius strictly smaller than 2^{-n}.

Proposition 1.2. Let Σ_1 and Σ_2 be finite alphabets, then $f : \Sigma_1 \to \Sigma_2$ is uniformly continuous iff for every recognisable language of Σ_2 L, $f^{-1}(L)$ is also recognisable.

Let $\widehat{\Sigma^*}$ denote the completion of Σ^*, whose points are called profinite words of Σ.

Proposition 1.3. Let M be a finite monoid and let $f : \Sigma^* \to M$ be uniformly continuous, then f uniquely extends to $\widehat{f} : \widehat{\Sigma^*} \to M$.

The following is the universal property of the free profinite monoid:

Proposition 1.4. Let Σ be a finite alphabet, then if $f : \Sigma \to M$ is a function to a finite monoid M, there exists a unique uniformly continuous monoid morphism $\widehat{f} : \widehat{\Sigma^*} \to M$ such that for each $a \in \Sigma$, $\widehat{f}(a) = f(a)$. $\widehat{f}$ is onto iff $f(\Sigma)$ generates M.

Proposition 1.5. Let Σ_1 and Σ_2 be finite alphabets, then every uniformly continuous morphism $f : \Sigma_1^* \to \Sigma_2^*$ uniquely extends to uniformly continuous $\widehat{f} : \widehat{\Sigma_1^*} \to \widehat{\Sigma_2^*}$.

The metric on $\widehat{\Sigma^*}$ is defined similarly. Let M be a finite monoid, then a morphism $f : \Sigma^* \twoheadrightarrow M$ separates profinite words $u, v \in \widehat{\Sigma^*}$ if $\widehat{f}(u) = \widehat{f}(v)$. For $u, v \in u, v \in \widehat{\Sigma^*}$, we set:

- $r(u, v) = \min\{|M| \mid M \text{ separates } u \text{ and } v\}$,
- $d(u, v) = 2^{-r(u,v)}$.

Proposition 1.6. The profinite topology on $\widehat{\Sigma^*}$ is the least topology making every continuous every morphism Σ^* onto a finite discrete monoid.

1.2 ω-terms

Let $x \in \widehat{\Sigma^*}$ be a profinite word, the profinite word x^ω is defined as

$$x^\omega = \lim_{n\to\infty} x^{n!}$$

The existence of x^ω follows from the following proposition:

Proposition 1.7. For each $x \in \widehat{\Sigma^*}$, the sequence $(x^{n!})_{n\in\mathbb{N}}$ is a Cauchy sequence converging to an idempotent element of $\widehat{\Sigma^*}$.

Note that x^ω is a choice of notation, but it is not an infinite word. One needs to intepret ω as the exponent of a finite semigroup. One can compute the image x^ω under a morphism to a finite monoid.

Let M be a finite monoid with exponent ω, let $f : \Sigma^* \to M$ and let $s = f(x)$ for $x \in \Sigma^*$. Then the sequence $s^{n!}$ converges to the idempotent s^ω of the subsemingroup of M generated by s. Therefore, one obtains

$$\hat{f}(x^\omega) = f(x)^\omega$$

Also, $(x^\omega)^\omega = x^\omega$ and $x^{\omega\omega} = x^\omega$. Besides, we have:

$$x^{\omega+1} = \lim_{n\to\infty} x^{n!+1}$$
$$x^{\omega-1} = \lim_{n\to\infty} x^{n!-1}$$

So we have by the definition:

$$xx^\omega = x^{\omega+1} = x^\omega x$$
$$xx^{\omega-1} = x^\omega = x^{\omega-1}x$$

One also gets $f(x^{\omega+1}) = s^{\omega+1}$, though the interpretation of $x^{\omega-1}$ is slightly more complicated. The minimal ideal of the subsemigroup generated by s is a group G with identity s^ω. But $s^{\omega-1}$ might not be in G. However, in M one has $\lim_{n\to\infty} s^{n!-1} = s^{2\omega-1}$, and, thus, $f(x^{\omega-1}) = s^{2\omega-1}$. Besides, $s^{2\omega-1}$ is the inverse of $s^{\omega+1}$ in G.

The ω-terms on Σ is the smallest submonoid of $\widehat{\Sigma^*}$ containing Σ^* and closed under operations $x \mapsto x^\omega$, $x \mapsto x^{\omega+1}$ and $x \mapsto x^{\omega-1}$.

1.3 Profinite Lambda Terms

One moves from profinite words to profinite lambda terms in the following way. For any finite set Q, one has the interpretation of any type A as the finite set $[\![A]\!]_Q$. Given another finite set Q' with a relation $R \subseteq Q \times Q'$, one can relate different interpretations of A by constructing a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$. Such relations are called *logical relations*. Lambda terms are parametric: that is, if M has type A and $R \subseteq Q \times Q'$, then $([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$.

Now let us discuss how one can define the exponential of relations. Let $S \subseteq P \times P'$ and $R \subseteq Q \times Q'$ be relations on finite sets, the exponential $S \Rightarrow R \subseteq (P \Rightarrow Q) \times (P' \Rightarrow Q')$ is given as:

$$S \Rightarrow R := \{(g,h) \mid \forall x \in P \ \forall y \in P' \ (xSy \to g(x)Rh(y))\}$$

Therefore, for any relation $R \subseteq Q \times Q'$ between finite sets Q and Q', one constructs a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ defined by induction on A as follows:

- $[\![p]\!] = R$,

- $[\![A \Rightarrow B]\!]_R = [\![A]\!]_R \Rightarrow [\![B]\!]_R$

The *fundamental lemma of logical relations* states that for any $M \in \Lambda_{\beta\eta}(A)$ for any A and for any $R \subseteq Q \times Q'$, the interpretations of M are related as follows:

$$([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$$

Further, we will be using partial surjections since they are stable under taking relational exponentials. Also, if $f : Q \twoheadrightarrow Q'$ is a partial surjection, so $[\![A]\!]_f \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ for any A. Besides, we have the inclusion of Boolean algebras $\mathrm{Reg}_{Q'}(A) \subseteq \mathrm{Reg}_Q(A)$, which is proved using the fundamental lemma of logical relations.

Theorem 1.1. The diagram of Boolean algebras $\mathrm{Reg}_{(.)}(A) : \mathbf{FinPSurj}^{op} \to \mathbf{BA}$:

$$(\mathrm{Reg}_{Q'}(A) \hookrightarrow \mathrm{Reg}_Q(A))_{f:Q \to Q' \in \mathbf{FinPSurj}}$$

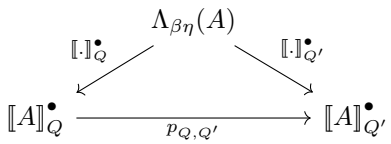
is directed, and its colimit in **BA** coincides with the Boolean algebra $\mathrm{Reg}(A)$ of regular languages for a higher-order type A.

Also, for any simple type A, the set of regular languages $\mathrm{Reg}(A)$ is closed under finite intersections.

Now let us discuss how the space of profinite lambda terms is constructed. The idea is that for any simple type A as parametric families of semantic elements, so they form a Stone space dual to the Boolean algebra $\mathrm{Reg}(A)$.

Proposition 1.8. Let Q, Q' be finite sets,

1. The set of atoms $\mathrm{At}_Q(A)$ of the finite Boolean algebra $\mathrm{Reg}_Q(A)$ is in bijection with the set $[\![A]\!]^\bullet_Q$ of definable elements of simple type A, given by the function $q \mapsto [\![.]\!]_Q^{-1}(\{q\}) : \mathrm{At}_Q(A) \to [\![A]\!]^\bullet_Q$.
2. For partial surjection $f : Q \twoheadrightarrow Q'$ and for any simple type A, the set $[\![A]\!]^\bullet_Q \subseteq [\![A]\!]_Q$ of definable elements is contained in the domain of $[\![A]\!]_f$, and the restriction of $[\![A]\!]_f$ to $[\![A]\!]^\bullet_Q$ is the unique function $p_{Q,Q'}$ such that the following triangle commutes:



Note that if $f, g : Q \to Q'$ are partial surjections, then, while the semantic interpretations $[\![A]\!]_f, [\![A]\!]_g : [\![A]\!]_Q \to [\![A]\!]'_Q$ are different, their restrictions to $[\![A]\!]^\bullet_Q$ are both equal to $p_{Q,Q'}$.

One defines profinite lambda terms of simple type A as follows.

Definition 1.1. Let A be a simple type, the set of *profinite lambda terms* is defined as the limit $\widehat{\Lambda}_{\beta\eta}(A)$ in **Set** of the diagram:

$$([\![A]\!]^\bullet_f : [\![A]\!]^\bullet_Q \twoheadrightarrow [\![A]\!]^\bullet_{Q'})_{f:Q \twoheadrightarrow Q' \in \mathbf{FinPSurj}}$$

In particular, by the definition of a limit in **Set**, it means that a profinite lambda term in $\widehat{\Lambda_{\beta\eta}(A)}$ is a family θ of definable terms $\theta_Q \in [\![A]\!]_Q$ for every finite set Q such that for every partial surjection

$f : Q \twoheadrightarrow Q'$ one has $[\![A]\!]^\bullet_f(\theta_Q) = \theta_{Q'}$. By Proposition 1.8, the above condition is equivalent to the condition that for every term $M : A$ for simple A and any finite sets Q and Q':

$$\theta_Q = [\![M]\!]_Q \wedge |Q'| \leq |Q| \Rightarrow \theta_{Q'} = [\![M]\!]_{Q'}$$

$\widehat{\Lambda_{\beta\eta}(A)}$ is equipped with a natural topology making it a Stone dual of the Boolean algebra $\mathrm{Reg}(A)$. One can define this topology as inherited from the inclusion

$$\widehat{\Lambda_{\beta\eta}(A)} \hookrightarrow \prod_Q [\![A]\!]^\bullet_Q$$

into the product space $\prod_Q [\![A]\!]^\bullet_Q$ where each $[\![A]\!]^\bullet_Q$ is considered a space with the discrete topology. Concretely, for any finite set Q and $q \in [\![A]\!]^\bullet_Q$, let $U_{Q,q}$ denote the set of profinite lambda terms taking value at Q:

$$U_{Q,q} := \{\theta \in \widehat{\Lambda_{\beta\eta}(A)} \mid \theta_Q = q\}$$

Now we take the collection $U_{Q,q}$ as the basis on $\widehat{\Lambda_{\beta\eta}(A)}$ for any Q and q.

The following is similar to [Geh16, Section 4.2]:

Proposition 1.9. The space $\widehat{\Lambda_{\beta\eta}(A)}$ is a Stone dual to the Boolean algebra $\mathrm{Reg}(A)$.

1.4 The cartesian closed category of profinite lambda terms

Let A be simple type, a *parametric family* is a family of points $\theta_Q \in [\![A]\!]_Q$ for a finite set Q, such that for any relation $R \subseteq Q \times Q'$ one has $\theta_Q [\![A]\!]_R \theta_{Q'}$. Every parametric family is preserved under partial surjections. Thus, any parametric family whose components are definable elements is a profinite term.

Theorem 1.2. Profinite terms $\theta \in \widehat{\Lambda}_{\beta\eta}(A)$ can be equivalently characterised as parametric families of definable terms $\theta_Q \in [\![A]\!]^\bullet_Q$.

Profinite lambda terms are assembled in a cartesian closed category by using the construction from [JM18].

Let $\mathcal{C}$ be a cartesian closed category quipped with a functor $\mathcal{P} : \mathcal{C} \to \mathbf{Set}$ with cartesian products preserved as follows. The canonical functions

$$(\mathcal{P}\pi_1, \mathcal{P}\pi_2) : \mathcal{P}(A \times B) \to \mathcal{P}A \times \mathcal{P}B$$
$$!_{\mathcal{P}\top} : \top \to \mathcal{P}\top$$

have the following inverses:

$$\mathsf{m}_{A,B} : \mathcal{P}A \times \mathcal{P}B \to \mathcal{P}(A \times B)$$
$$\mathsf{m}_\top : \mathcal{P}\top \to \top$$

Let us define the category $\mathcal{C}[\mathcal{P}]$ whose objects are the objects of $\mathcal{C}$ and whose hom-sets are defined as

$$\mathcal{C}[\mathcal{P}](A, B) = \mathcal{P}(B^A)$$

Proposition 1.10. The category $\mathcal{C}[\mathcal{P}]$ is cartesian closed and comes equipped with the identity-on-object (cartesian closed) functor $\mathbf{idonobj}_{\mathcal{C}[\mathcal{P}]} : \mathcal{C} \to \mathcal{C}[\mathcal{P}]$ strictly preserving exponentials and cartesian products.

Let us recall how the cartesian closed category **Lam** of simply typed lambda terms is generated by the terminal category.

Definition 1.2. The category **Lam** has objects as simple types and hom-sets $\mathbf{Lam}(A, B)$ are defined as $\Lambda_{\beta\eta}(A \Rightarrow B)$ for simple types A and B.

Let us consider the functor $\mathcal{P} : \mathbf{Lam} \to \mathbf{Set}$ defined as $\mathcal{P}(A) = \widehat{\Lambda_{\beta\eta}}(A)$. $\mathcal{P}$ sends every term $M : A \Rightarrow B$ to the set-theoretic function sending a profinite term θ of type A on the family $(\llbracket M \rrbracket_Q(\theta_Q))$ that can be shown to be profinite term of type B via the fundamental lemma of logical relations. One has

$$\mathcal{P}(A \times B) \cong \mathcal{P}(A) \times \mathcal{P}(B)$$
$$\mathcal{P}\top \cong \top$$

Therefore, we obtain a cartesian closed category $\mathbf{ProLam} := \mathbf{Lam}(\mathcal{P})$ whose hom-sets are $\mathbf{ProLam}(A, B) := \widehat{\Lambda_{\beta\eta}}(A \Rightarrow B)$.

The functor $\mathcal{P}$ is valued in **Set**, but it also can be value in any cartesian category $\mathcal{S}$ as long as $\mathcal{P}$ isomorphically preserves cartesian products, we will get the cartesian category $\mathcal{C}[\mathcal{P}]$ enriched over $\mathcal{S}$. In particular, the functor $\mathcal{P} : \mathbf{Lam} \to \mathbf{Set}$ factors through the category of Stone spaces as follows:

$$\mathbf{Lam} \xrightarrow{\widehat{\Lambda}_{\beta\eta}(.)} \mathbf{Stone} \xrightarrow{\text{forget}} \mathbf{Set}$$

The category **ProLam** is equipped with a cartesian closed idenity-on-functor functor $\mathbf{idonobj} : \mathbf{Lam} \to \mathbf{ProLam}$ derived from the fact that **Lam** is a free cartesian closed category.

Proposition 1.11. The functor $\mathbf{idonobj} : \mathbf{Lam} \to \mathbf{ProLam}$ is faithful.

Proof. The category **ProLam** can be obtained as the limit of a codirected diagram of cartesian closed category described as follows. Let Q be a finite set and let A be a simple type, consider the following equivalence relation:

$$\sim_Q^A \subseteq \Lambda_{\beta\eta}(A) \times \Lambda_{\beta\eta}(A)$$

on the set of terms of type A modulo $\beta\eta$-equivalence defined as

$$M \sim_Q^A N \leftrightarrow \llbracket M \rrbracket_Q = \llbracket N \rrbracket_Q$$

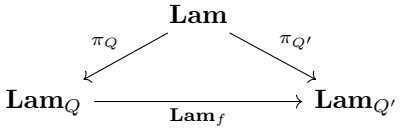
Fix a finite set Q, the family $\sim_Q$ is parametrised by simple types defines a congruence relation on the category **Lam**:

$$f \sim_Q^{A \Rightarrow B} f' \wedge g \sim_Q^{B \Rightarrow C} g' \Rightarrow g \circ f \sim_Q^{A \Rightarrow C} g' \circ f'$$

Therefore we obtain the category $\mathbf{Lam}_Q := \mathbf{Lam}/Q$ with morphisms defined as:

$$\mathbf{Lam}_Q(A, B) = \Lambda_{\beta\eta}(A \Rightarrow B)/\sim_Q^{A \Rightarrow B}$$

First of all, one can observe that for any finite set Q, $\mathbf{Lam}_Q$ is cartesian closed and equipped with cartesian closed identity-on-functor $\pi_Q : \mathbf{Lam} \to \mathbf{Lam}_Q$. Besides, every partial surjection $f : Q \twoheadrightarrow Q'$ in **FinPSurj** induces a cartesian closed identity-on-functor $\mathbf{Lam}_f : \mathbf{Lam}_Q \to \mathbf{Lam}_{Q'}$ such that the following triangle commutes:



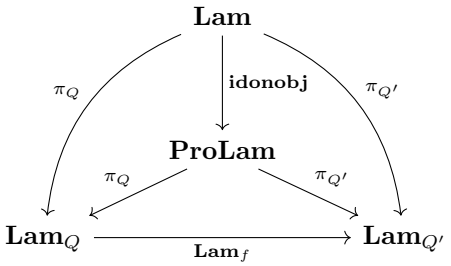
Further, one observes that the category **ProLam** is the codirected limit of the diagram of cartesian closed categories $\mathbf{Lam}_Q$ indexed by finite sets and partial surjections between them. The projection functor $\pi_Q : \mathbf{ProLam} \to \mathbf{Lam}_Q$ is defined by transporting every morphism $\theta : A \to B$ defined as a family θ of definable elements in

$$\mathbf{ProLam}(A, B) = \widehat{\Lambda}_{\beta\eta}(A \Rightarrow B)$$

to the corresponding θ_Q in

$$\mathbf{Lam}_Q(A \Rightarrow B) = \widehat{\Lambda}_{\beta\eta}(A \Rightarrow B)/_Q^{A \Rightarrow B} \cong [\![A \Rightarrow B]\!]_Q^\bullet$$

so the canonical functor makes the following diagram commute:



Further, one concludes that **idonobj** is faithful as follows. By Statman's finite completeness theorem [Sta82], for every pair of terms $M, N : A \to B$, then either $M \equiv_{\beta\eta} N$ or there is a finite set Q such that the interpretations $[\![M]\!]_Q = \pi_Q(M)$ and $[\![N]\!]_Q = \pi_Q(N)$ are different. In this case $\mathbf{idonobj}(M)$ and $\mathbf{idonobj}(N)$ are different as well, and therefore **idonobj** is faithful. □

Further, let us recall some facts from the higher-order language theory.

Proposition 1.12. Let Σ be a finite alphabet, then the Boolean algebras $\mathrm{Reg}(\mathrm{Church}_\Sigma)$ and $\mathrm{Reg}(\Sigma)$ given by the Church encoding.

Proof. By the Church encoding, there is a one-to-one correspondence between $L \subseteq \Sigma^*$ and $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$. One needs to show that $L \subseteq \Sigma^*$ is regular iff the associated element $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$ belongs to $\mathrm{Reg}(\mathrm{Church}_\Sigma)$.

Let L be a language recognised by a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, \mathrm{Acc})$. The associated set $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$ is defined by the inverse image of the semantic bracket $[\![.]\!]_Q : \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q$ of the set of functionals in $[\![\mathrm{Church}_\Sigma]\!]_Q$ defined in the following way:

$$\mathrm{eval}^{-1}_{\delta,q_0}(\mathrm{Acc}) = \{F \in [\![\mathrm{Church}_\Sigma]\!]_Q \mid F(\delta_{a_1}, \ldots, \delta_{a_n})(q_0) \in \mathrm{Acc}\}$$

Therefore $\mathcal{L}$ is an element of $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$, and therefore, an element of $\mathrm{Reg}(\mathrm{Church}_\Sigma)$.

Conversely, by definition of $\mathrm{Reg}(\mathrm{Church}_\Sigma)$, it suffices to show that for every finite set Q, a subset $\mathcal{L} \in \mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ corresponds to some regular language $L \subseteq \Sigma^*$. We can suppose that $\mathcal{L}$ has the following form

$$\mathcal{L} = [\![.]\!]_Q^{-1}(\{F\}) = \{M \in \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \mid [\![M]\!]_Q = F\}$$

where F is a functional in $[\![\mathrm{Church}_\Sigma]\!]_Q$. The corresponding set $L \subseteq \Sigma^*$ is the finite intersection of regular languages recognised by the deterministic finite automata of the form $\mathcal{A} = (Q, \delta, q_0, \{q_f\})$ where the unique finite state q_f is equal to $q_f = F(\delta_{a_1}, \ldots, \delta_{a_n})(q_0)$, therefore the set L is a regular language itself being the finite intersection of regular languages. □

By Stone duality and the above theorem, one also has that for every finite alphabet Σ, there is a homeomorphism $\widehat{\Lambda}_{\beta\eta}(\mathrm{Church}_\Sigma) \cong \widehat{\Sigma^*}$ between the space $\widehat{\Lambda}_{\beta\eta}(\mathrm{Church}_\Sigma)$ of profinite lambda terms Church_Σ and the space of profinite words $\widehat{\Sigma^*}$.

This is how the concept of an ω-word transferred to lambda terms.

Proposition 1.13. Let A be a simple type, then there exists a term $\Omega_A : (A \Rightarrow A) \Rightarrow A \Rightarrow A$ which, for any profinite term $M \in \widehat{\Lambda}_{\beta\eta}(A \Rightarrow A)$ satisfies the idempotency equation:

$$(\Omega_A M) \circ (\Omega_A M) = \Omega_A M$$

2 h-adic topology

The following summary is based on [Kas12, Chapter XVI].

Consider the complex algebra $K = \mathbb{C}[[h]]$ of complex formal series in one variable h. Every element $f \in \mathcal{K}$ has the form

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where $a_n \in \mathbb{C}$ for each $n \in \mathbb{N}$. Every polynomial $f \in \mathbb{C}[h]$ is clearly a formal series. K form a commutative ring with operations given as

- $f + g = \sum\limits_{n=0}^{\infty} a_n h^n + \sum\limits_{n=0}^{\infty} b_n h^n = \sum\limits_{n=0}^{\infty} (a_n + b_n) h^n$
- $f \cdot g = (\sum\limits_{n=0}^{\infty} a_n h^n) \cdot (\sum\limits_{n=0}^{\infty} b_n h^n) = \sum\limits_{n=0}^{\infty} (\sum\limits_{n=p+q} a_p b_q) h^n$

A formal series f is invertible iff its every coefficient is non-zero.

For any $n > 0$, consider the algebra $K_n = \mathbb{C}[h]/(h^n)$ of truncated polynomials obtained as the quotient of the ring of complex polynomials generated by the ideal (h^n). There is a morphism

$$\pi_n : K \to K_n$$

sending each formal series $\sum\limits_{n=0}^{\infty} a_n h^n$ to the class of $\sum\limits_{k=0}^{n-1} a_k h^k$ modulo (h^n). π_n is onto and its kernel is the ideal $h^n K$ generated by h^n in the ring of formal series. Therefore, we have the following isomorphism:

$$\mathbb{C}[[h]]/(h^n) \cong \mathbb{C}[h]/(h^n)$$

For $n > 0$, there is a surjective map of algebras $p_n : K_n \to K_{n-1}$ generated by the inclusion of ideals $(h^n) \subset (h^{n-1})$. So we have the inverse system of algebras $(K_n, p_n)_{n>0}$ and its inverse limit $\varprojlim\limits_n K_n$. One has $p_n \circ \pi_n = \pi_{n-1}$ for each n. Also, there is a unique morphism of algebras from K to $\varprojlim\limits_n K_n$ whose composition with the projection of the inverse limit is equal to π_n. Besides, the map $\pi : \mathbb{C}[[h]] \twoheadrightarrow \varprojlim\limits_n \mathbb{C}[[h]]/(h^n)$ is the isomorphism, which is partly the instance of Birkhoff's subdirect representation theorem, [SB81, Theorem 8.6].

Let us equip K with the *h-adic topology*. The family of of subsets $\pi_n^{-1}(U_n)$ for $n > 0$ and $U_n \subseteq K_n$ form an open basis on K. $\{0\}$ is a family of open neighbourhoods in the discrete set K_n, the family $\pi_n^{-1}(0)$ is a family of open neighbourhoods of 0 in K for the h-adic topology. There is

also the structure of a metric space on K given as follows. Given a formal series $f = \sum_{n=0}^{\infty} a_n h^n$, let $o(f)$ denote the unique natural number such that $a_{o(f)} \neq 0$ and $a_k = 0$ for each $k < o(f)$. For $f = 0$, set $o(f) = +\infty$ and extend $(\mathbb{N}, <)$ by adding the top element $+\infty$. It is easy to see that:

$$(h^n) = \{f \in K \mid o(f) > n - 1\}$$

and therefore

$$\bigcap_{n>0} (h^n) = \{0\}$$

We also have for any series $f, g \in K$:

$$o(f + g) \geq \min(o(f), o(g))$$

Now define $|.| : K \to \mathbb{R}_{\geq 0}$ as

$$|f| = 2^{-o(f)}$$

Then if we put $d(f, g) = |f - g|$, then $(\mathbb{C}[[h]], d)$ becomes an ultrametric space.

Let M be a left K-module, consider the family $(h^n M)_{n>0}$ of submodules of M with the canonical K-linear projections:

$$p_n : M_n = M/h^n M \to M_{n-1} = M/h^{n-1} M$$

They form an inverse system of K-modules, so we can consider the inverse limit

$$\widetilde{M} = \varprojlim_n M_n$$

having a natural structure of a left K-module. $\widetilde{M}$ has a natural topology as well, the inverse limit topology, for which the family of submodules $(h^n \widetilde{M})_n$ form an open basis. The projections $i_n : M \to M_n$ induce a unique K-linear map $i : M \to \widetilde{M}$ such that $\pi_n \circ i = i_n$ for each n. The kernel of i is given as

$$\mathrm{Ker}(i) = \bigcap_{n>0} h^n M$$

A K-module M is *separated* if $\mathrm{Ker}(i) = \{0\}$. It is *complete* if i is injective.

For any module K, the module $M/\mathrm{Ker}(i)$ is separated and the completion $\widetilde{M}$ of M is complete.

Generally, any separated complete K-module M is equipped with the h-adic topology induced by the inverse limit topology on $\widetilde{M}$ via $M \cong \widetilde{M}$.

Now let us describe the following class of separated complete K-modules. Let V be a vector space over $\mathbb{C}$, let $V[[h]]$ denote the set of all formal series in variable h

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where a's range over elements of V. There is a left K-module on $V[[h]]$, and such modules are called *topologically free modules*. One can observe that any topologically free module is separated and complete. Also, if $\{e_i\}_{i \in I}$ is a basis of V, then the K-submodule generated by $\{e_i\}_{i \in I}$ is dense in V for the h-adic topology. Besides, let N be a complete separated K-module, there is a natural isomorphism in the category of K-modules and K-linear maps:

$$\mathrm{Hom}_K(V[[h]], N) \cong \mathrm{Hom}_K(V, N)$$

Further, a K-module M is topologically free iff it is complete, separated and torsion-free, that is, there is such that $mh \neq 0$ for any non-zero $m \in M$.

Let M and N be left modules over the algebra $K = \mathbb{C}[[h]]$. The K-module $M \otimes_K N$ is the quotient of the vector space $M \otimes N$ obtained by spanning all elements of the form $fm \otimes n - m \otimes fn$ for $f \in K$, $m \in M$ and $n \in N$.

Definition 2.1. The topological tensor product $M\widetilde{\otimes}N$ of M and N is the h-adic completion of $M \otimes_K N$:

$$M\widetilde{\otimes}N := \widetilde{M \otimes_K N} = \varprojlim_{n>0}(M \otimes_K N)/h^n(M \otimes_K N)$$

By definition, $M\widetilde{\otimes}N$ is complete.

Take $m \in M$ and $n \in N$, let $m\widetilde{\otimes}n$ denote the image of $m \otimes n$ under the following image:

$$M \otimes N \to M \otimes_K N \to M\widetilde{\otimes}N$$

The subspace of the topological tensor product spanned by all elements of this form is dense in $M \otimes N$.

Proposition 2.1.

1. The class left modules over the algebra $K = \mathbb{C}[[h]]$ with $\widetilde{\otimes}$ is endowed with the structure of a symmetric monoidal category
2. If M and N be topologically free modules, so is $M\widetilde{\otimes}N$, that is:

$$V[[h]]\widetilde{\otimes}W[[h]] = (V \otimes W)[[h]].$$

2.1 Topological Algebras

A *topological algebra* is a triple (A, μ, η) where A is a module over the ring $K = \mathbb{C}[[h]]$ and $\mu : A\widetilde{\otimes}A \to A$ and $\eta : A \to K$ are k-linear maps such that:

- $\mu \circ (1_A\widetilde{\otimes}\mu) = \mu \circ (\mu\widetilde{\otimes}1_A)$,
- $\mu \circ (\eta\widetilde{\otimes}1_A) = 1_A = \mu \circ (1_A\widetilde{\otimes}\eta)$.

For $a, a' \in A$, aa' will stand for $\mu(a\widetilde{\otimes}a')$.

Let (A, μ, η) be a topological algebra and let $f(h) = \sum c_n h^n$ be a formal series with complex coefficients. For a, the formula

$$f(ha) = \sum_{n\geq 0} c_n a^n h^n$$

defines a unique element in the inverse limit $\widetilde{A} = \varprojlim_n A/h^n A$. If A is separated and complete, then it defines an element $f(ha)$ in $A \cong \widetilde{A}$. This procedure can be applied to the exponential function $e^h = \sum_{n\geq 0} \frac{h^n}{n!}$ yeilding elements of the form

$$e^{ha} = \sum_{n\geq 0} \frac{a^n h^n}{n!}$$

in any separate complete topological algebra. If $a' \in A$, then a' *commutes* with a whenever

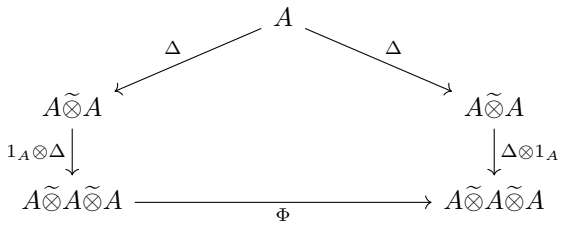
$$e^{a+a'} = e^{ha}e^{ha'}$$

e^{ha} is also invertible in A with e^{-ha}.

A morphism $f : (A, \mu, \eta) \to (A', \mu', \eta')$ of topological algebras is a K-linear map $f : A \to A'$ such that $f \circ \mu = \mu' \circ (f \widetilde{\otimes} f)$ and $f \circ \eta = \eta'$.

A topological quasi-bialgebra is a structure $(A, \mu, \eta, \Delta, \varepsilon, \Phi)$, where (A, μ, η) is a topological algebra, $\Delta : A \to A\widetilde{\otimes}A$ and $\varepsilon : A \to K$ are K-linear maps and $\Phi : A^{\widetilde{\otimes}3} \cong A^{\widetilde{\otimes}3}$:

- The following commutes:



- (A, μ, η) is a topological coalgebra,
- $(1_A\widetilde{\otimes}1_A\widetilde{\otimes}\Delta)(\Phi)(\Delta\widetilde{\otimes}1_A\widetilde{\otimes}1_A)(\Phi) = \Phi_{234}(1_A\widetilde{\otimes}\Delta\widetilde{\otimes}1_A)(\Phi)\Phi_{123}$,
- $(1_A\widetilde{\otimes}\varepsilon\widetilde{\otimes}1_A) = 1 \otimes 1$.

A topological quasi-bialgebra is a topological bialgebra whenever $\Phi = 1^{\widetilde{\otimes}3}$.

A topological quasi-bialgebra $(A, \mu, \eta, \Delta, \varepsilon, \Phi)$ is a *topological braided quasi-algebra* if there is an invertible element $R \in A\widetilde{\otimes}A$ such that

- $\Delta^{op}(a) = R\Delta(a)R^{-1}$ for any $a \in A$,
- $(1_A\widetilde{\otimes}\Delta)(R) = (\Phi_{231})^{-1}R_{13}\Phi_{213}R_{12}(\Phi_{123})^{-1}$
- $(\Delta\widetilde{\otimes}1_A)(R) = \Phi_{321}R_{13}(\Phi_{132})^{-1}R_{23}\Phi_{123}$

R is called the universal R-matrix of A.

A *gauge transformation* on a topological quasi-algebra is an invertible element $F \in A\widetilde{\otimes}A$ such that

$$(1_A\widetilde{\otimes}\varepsilon_A)F = (\varepsilon_A\widetilde{\otimes}1_A)F = 1$$

Example 2.1. Let $A_0 = (A_0, \mu_0, \eta_0, \Delta_0, \varepsilon_0, \Phi_0, R_0)$ be a braided quasi-bialgebra over $\mathbb{C}$. One can define the following topological quasi-braided algebra:

$$A_0[[h]] = (A_0[[h]], \mu, \eta, \Delta, \varepsilon, \Phi_0, R_0)$$

on the space of formal series with coefficients in A_0 where $\mu, \eta, \Delta, \varepsilon$ are unique K-linear maps such that

$$\forall f \in K \;\; \eta(f) = f1 = f\eta_0 1$$

and for all $a, a' \in A$

- $\mu(a\widetilde{\otimes}a') = \mu_0(a \otimes a')$,

- $\Delta(a) = \Delta_0(a)$,
- $\varepsilon(a) = \varepsilon_0(a)$.

$A_0[[h]]$ is called the trivial topological braided quasi-algebra associated to A_0.

Example 2.2. Let $A = (A, \mu, \eta, \Delta, \varepsilon, \Phi, R)$ be a topological braided quasi-algebra. We have $(A\widetilde{\otimes}A)/h(A\widetilde{\otimes}A) \cong A/hA \otimes A/hA$, so the K-linear maps $\mu, \eta, \Delta, \varepsilon$ induce $\mathbb{C}$-linear maps:

- $\overline{\mu} : A/hA \otimes A/hA \to A/hA$,
- $\overline{\Delta} : A/hA \to A/hA \otimes A/hA$,
- $\overline{\eta} : \mathbb{C} \to A/hA$,
- $\overline{\varepsilon} : A/hA \to \mathbb{C}$.

Define $\overline{\Phi}$ as the class of Φ modulo $(A/hA)^{\otimes 3}$ and define $\overline{R}$ as the class of R modulo $(A/hA)^{\otimes 3}$, then $\overline{A} = (\overline{A}, \overline{\mu}, \overline{\eta}, \overline{\Delta}, \overline{\varepsilon}, \overline{\Phi}, \overline{R})$ is a braided quasi-algebra.

A *topological A-module* M over a topological algebra (A, μ, η) is a left K-module with a K-linear map $\mu_M : A\widetilde{\otimes}M \to M$ such that

$$\begin{array}{ccc} A \otimes A \otimes M & \xrightarrow{\mu\widetilde{\otimes}1_M} & A\widetilde{\otimes}M \\ {\scriptstyle 1_A\widetilde{\otimes}\mu_M}\downarrow & & \downarrow{\scriptstyle \mu_M} \\ A \otimes M & \xrightarrow[\mu_M]{} & M \end{array} \qquad \begin{array}{ccc} M & \xrightarrow{\eta\widetilde{\otimes}1_M} & A \otimes M \\ & \searrow & \downarrow{\scriptstyle \mu_M} \\ & & M \end{array}$$

One writes $\mu_M(a\widetilde{\otimes}m) := am$ for $a \in A$ and $m \in M$. Let M and N be topological A-modules, then their tensor product $M\widetilde{\otimes}N$ is a topological $A\widetilde{\otimes}A$-module. If A has a comultiplication $\Delta : A \to A\widetilde{\otimes}A$. One can pull back the $A\widetilde{\otimes}A$-module structure on $M\widetilde{\otimes}N$ to a topological A-module structure given as follows, for $a \in A$, $m \in M$ and $n \in N$:

$$a(m\widetilde{\otimes}n) = \Delta(a)(m\widetilde{\otimes}n)$$

Let A be a topological braided algebra with universal R-matrix R, then for any topological A-module M the k-linear automorphism $c^R_{M,M}$ is defined by

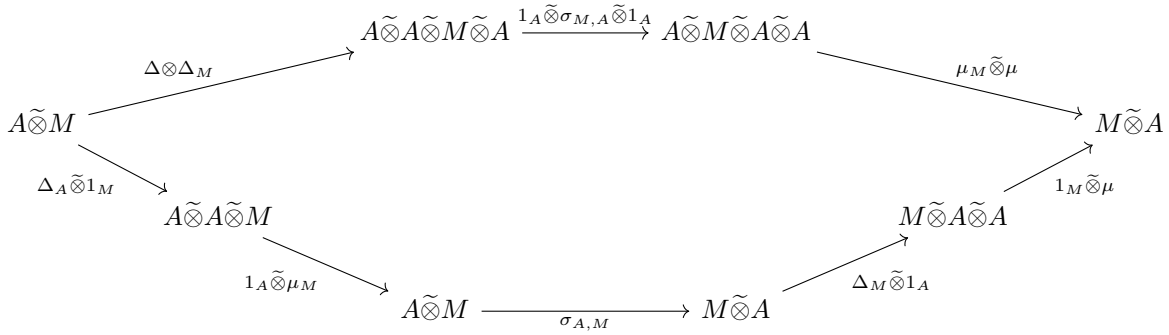
$$c^R_{M,M}(m_1\widetilde{\otimes}m_2) = (R(m_1\widetilde{\otimes}m_2))_{21}$$

is a solution of the Yang-Baxter equation in $M^{\widetilde{\otimes}3}$.

The universal R-matrix of a topological braided quasi-bialgebra gives rise to a representation of of the braid group $\mathcal{B}_n$ on the topological A-module $M^{\widetilde{\otimes}n}$ for each $n > 1$ and a topological A-module M.

A topological A-module M over a topological bialgebra $A = (A, \mu, \eta, \Delta, \varepsilon)$ equipped with a left action $\mu_M : A\widetilde{\otimes}M \to M$ is a *topological crossed A-module* if there is a k-linear map $\Delta_M : M \to M\widetilde{\otimes}A$ such that:

$$\begin{array}{ccc} M & \xrightarrow{\Delta_A} & M\widetilde{\otimes}A \\ {\scriptstyle \Delta_A}\downarrow & & \downarrow{\scriptstyle \Delta_M\widetilde{\otimes}A} \\ M\widetilde{\otimes}A & \xrightarrow[1_M\widetilde{\otimes}\Delta]{} & M\widetilde{\otimes}A\widetilde{\otimes}A \end{array} \qquad \begin{array}{ccc} M & \xrightarrow{\Delta_A} & M\widetilde{\otimes}A \\ & \searrow & \downarrow{\scriptstyle 1_M\widetilde{\otimes}\varepsilon} \\ & & M \end{array}$$



2.2 Quantum Enveloping Algebras

Definition 2.2. A *complex Lie algebra* is a vector space $\mathfrak{g}$ over $\mathbb{C}$ with the operation $[.,.] : \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g}$ satisfying the following axioms for all $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{C}$:

1. $[x, x] = 0$,
2. $[x, [y, z]] + [y, [x, z]] + [z, [x, y]] = 0$,
3. $[ax + by, z] = a[x, z] + b[y, z]$,
4. $[z, ax + by] = a[z, x] + b[z, y]$.

Any complex Lie algebra $\mathfrak{g}$ has a universal enveloping algebra $U(\mathfrak{g})$ equipped with the structure of bialgebra as follows for all $x \in U(\mathfrak{g})$:

- $\varepsilon(x) = 0$,
- $\Delta(x) = 1 \otimes x + x \otimes 1$

$U(\mathfrak{g})$ also has a trivial braided quasi-bialgebra structure with $\Phi = 1 \otimes 1 \otimes 1$ and $R = 1 \otimes 1$. By the Poincaré–Birkhoff–Witt theorem, every Lie algebra $\mathfrak{g}$ embeds into $U(\mathfrak{g})$.

Definition 2.3. A *quantum enveloping algebra* (QUE) for the Lie algebra $\mathfrak{g}$ is a topological braided quasi-bialgebra $(A, \mu, \eta, \Delta, \varepsilon, \Phi, R)$ such that A is a topologically free module, and the induced braided quasi-algebra $\overline{A} = (\overline{A}, \overline{\mu}, \overline{\eta}, \overline{\Delta}, \overline{\varepsilon}, \overline{\Phi}, \overline{R})$ as in Example 2.2 coincides with $U(\mathfrak{g})$ and η coincides with $\overline{\eta}$.

First of all, a QUE is topologically free, so $A = (A/hA)[[h]]$ as a left K-module. Therefore $A/hA = U(\mathfrak{g})$. So $A = U(\mathfrak{g})[[h]]$ as a K-module. So we derive from Proposition 2.1:

$$\forall n > 0 \;\; A^{\widetilde{n}} = (U(\mathfrak{g})^{\otimes n})[[h]]$$

Further, maps μ, η, Δ and ε are determined by their restrictions to $U(\mathfrak{g})\otimes U(\mathfrak{g})$, $\mathbb{C}$, $U(\mathfrak{g})$ and $U(\mathfrak{g})$ respectively. For $a, a' \in U(\mathfrak{g})$ we have

$$\mu(a \otimes a') = \sum_{n\geq 0} \mu_n(a \otimes a')h^n$$

where $\{\mu_n : U(\mathfrak{g})\otimes U(\mathfrak{g}) \to U(\mathfrak{g}) \,|\, n \geq 0\}$ is a family of linear maps such that μ_0 is the multiplication in the enveloping algebra. Similarly we have:

$$\forall a \in U(\mathfrak{g})\ \ \Delta(a) = \sum_{n\geq 0} \Delta_n(a)h^n$$
$$\forall a \in U(\mathfrak{g})\ \ \sum_{n\geq 0} \varepsilon_n(a)h^n$$
$$\forall f \in \mathbb{C}[[h]]\ \ \eta(f) = f1$$

By Proposition 2.1, we also write:

$$\Phi = \sum_{n\geq 0} \Phi_n h^n$$
$$R = \sum_{n\geq 0} R_n h^n$$

By definition, a quantum enveloping algebra A is associated to a Lie algebra $\mathfrak{g}$. $\mathfrak{g}$ is recovered from A as

$$\mathfrak{g} = \{x \in A/hA \mid \Delta_0(a) = 1 \otimes a + a \otimes 1\}$$

3 Duality for Topological $[[h]]$-modules and topological Hopf algebras

This section is loosely based on [Gav02].

Fix a field k, so $k[[h]]$ is by default a topological ring, that is, it is equipped with the h-adic topology. Let X be any $k[[h]]$-module, then we set

$$X_0 := X/hX = k \otimes_{k[[h]]} X,$$

a k-module via scalar restriction

$$k[[h]] \to k[[h]]/hk[[h]] \cong k$$

that we call *the specialisation of X at $h = 0$*, or, *the semiclassical limit of X*. Let

$$X \xrightarrow{h\to 0} \overline{Y}$$

We have discussed that if X is complete, torsion-less and separated, then there is a natural isomorphism of k-modules $X \cong X_0[[h]]$.

Let we set ${}^F X := k((h)) \otimes_{k[[h]]} X$, a vector space over $k((h))$ without any topology.

Let X be a topological $k[[h]]$-module, we let:

- $X^* := \mathrm{Hom}_{k[[h]]}(X, k[[h]])$, the *full dual* of X,
- $X^\star := \{f \in X^* \mid f \text{ is continuous}\}$, the *topological dual* of X.

There are two monoidal categories of topological k-modules:

- $\mathcal{T}_{\widehat{\otimes}}$, which is modelled on the monoidal category of discrete topological k-vector spaces k-vector spaces.
- $\mathcal{P}_{\widetilde{\otimes}}$, which is modelled on the monoidal category of linearly compact topological k-vector spaces.

The monoidal category $\mathcal{T}_{\widehat{\otimes}}$ consists of all topologically free modules (that is, modules isomorphic to $V[[h]]$ for some k-vector space V) and $k[[h]]$-linear maps. The tensor product $T_1 \widehat{\otimes} T_2$ is the separated h-adic completion of the algebraic tensor product $T_1 \otimes_{k[[h]]} T_2$ for $T_1, T_2 \in \mathcal{T}_{\widehat{\otimes}}$.

The monoidal category $\mathcal{P}_{\widetilde{\otimes}}$ is the closure under isomorphism of the class of topological modules of type $k[[h]]^E$ (the cartesian product indexed by E with the Tikhonov product topology) for some set E. These modules are complete w.r.t the weak topology. They are isomorphic to the projective limit of their finite free submodule (with the h-adic topology). Morphisms in $\mathcal{P}_{\widetilde{\otimes}}$ are $k[[h]]$-linear continuous maps. The tensor product $P_1 \widetilde{\otimes} P_2$ for $P_1, P_2 \in \mathcal{P}_{\widetilde{\otimes}}$ is the completion of $P_1 \otimes_{k[[h]]} P_2$ w.r.t. the weak topology: thus, if $P_i \cong k[[h]]^{E_i}$ for $i = 1, 2$, then one has $P_1 \widetilde{\otimes} P_2 \cong k[[h]]^{E_1 \times E_2}$.

Note that objects in both $\mathcal{T}_{\widehat{\otimes}}$ and $\mathcal{P}_{\widetilde{\otimes}}$ are complete w.r.t. the h-adic topology.

Further, $\mathcal{HA}_{\widehat{\otimes}}$ is the subcategory of $\mathcal{T}_{\widehat{\otimes}}$ whose objects are Hopf algebras in $\mathcal{T}_{\widehat{\otimes}}$ and whose morphisms are Hopf algebra morphisms in $\mathcal{T}_{\widehat{\otimes}}$. Similarly, $\mathcal{HA}_{\widetilde{\otimes}}$ is the subcategory of Hopf algebras from $\mathcal{P}_{\widetilde{\otimes}}$ whose morphisms are Hopf algebra morphisms in $\mathcal{P}_{\widetilde{\otimes}}$. $\mathcal{HA}_{\widetilde{\otimes}}^{\omega - I}$ is the full subcategory of $\mathcal{HA}_{\widetilde{\otimes}}$ consisting of all $H \in \mathcal{HA}_{\widetilde{\otimes}}$ whose weak topology coincides with the I_H-adic topology where

$$I_H := hH + \mathrm{Ker}(\varepsilon) = \varepsilon^{-1}(hk[[h]])$$

Note that objects in $\mathcal{HA}_{\widehat{\otimes}}$ and $\mathcal{P}_{\widetilde{\otimes}}$ are *topological Hopf algebras*, in particular, the comultiplication in those algebras is given as:

$$\Delta(x) = \sum_{(x)} x_{(1)} \otimes x_{(2)}$$

4 On Dialogue Categories and their internal language

Definition 4.1. A dialogue category is a monoidal category $(\mathcal{C}, \otimes, \mathbb{1})$ equipped with an object $\perp$ along with a family of isomorphisms

$$\varphi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(B, A \multimap \perp)$$

natural in x and y; along with a family of isomorphisms

$$\psi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(A, \perp \circ\!\!- B)$$

natural in A and B as well.

4.1 Term Assignment for Tensorial Logic

Figure 1: Term Calculus for Tensorial Logic

$$\frac{}{\vdash \mathbf{1} : \mathbb{1}}\ \mathbb{1}\mathbf{I} \qquad \frac{}{x : A \vdash x : A}\ \mathbf{ax} \qquad \frac{\Gamma \vdash M : \mathbb{1} \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N : B}\ \mathbb{1}\mathbf{E}$$

$$\frac{x : A, \Gamma \vdash M : \bot}{\Gamma \vdash \nabla_l x.M : A \multimap \bot}\ \multimap \bot_{\mathbf{I}} \qquad \frac{\Delta \vdash N : A \qquad \Gamma \vdash M : A \multimap \bot}{\Gamma, \Delta \vdash N \triangleleft M : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma, x : A \vdash M : \bot}{\Gamma \vdash \nabla_r x.M : \bot \circ\!\!- A}\ \bot \circ\!\!-_{\mathbf{I}} \qquad \frac{\Gamma \vdash M : A \circ\!\!- \bot \qquad \Delta \vdash N : A}{\Gamma, \Delta \vdash M \triangleright N : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma \vdash M : A \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash M \otimes N : A \otimes B}\ \otimes\mathbf{I} \qquad \frac{\Gamma \vdash M : A \otimes B \qquad \Delta, x : A, y : B \vdash N : C}{\Delta, \Gamma \vdash \mathbf{let}\ x \otimes y\ =\ M\ \mathbf{in}\ N : C}\ \otimes\mathbf{E}$$

Definition 4.2. Reduction rules for tensorial logic:

$(\multimap \bot\beta)$ $N \triangleleft (\nabla_l x.M) \to M[x := N]$,

$(\bot \circ\!\!- \beta)$ $(\nabla_r x.M) \triangleright N \to M[x := N]$,

$(\mathbb{1}\beta)$ $\mathbf{let}\ \mathbf{1} = \mathbf{1}\ \mathbf{in}\ N \to N$,

$(\otimes\beta)$ $\mathbf{let}\ x \otimes y = M \otimes N\ \mathbf{in}\ P \to P[x := M, y := N]$,

$(\multimap \bot\eta)$ $(\nabla_l x.x \triangleleft M) \to M$ if x is not free in M,

$(\bot \circ\!\!- \eta)$ $(\nabla_r x.M \triangleright x) \to M$ if x is not free in M,

$(\otimes\eta)$ $\mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N[z := x \otimes y] \to N[z := M]$,

$(\otimes_{\mathbf{nat}})$ $P[w := \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N] \to \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ P[w := N]$,

$(\mathbb{1}_{\mathbf{nat}})$ $P[w := \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N] \to \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ P[w := N]$.

In non-commutative tensorial logic, contexts in antecedents are lists, but in the formulation with antecedents as multisets, then we can just unify left and right negations.

4.2 Sums

Finite sums are given by the following inference rules.

$$\frac{\Gamma \vdash M : A}{\Gamma \vdash \mathbf{inl}(M) : A \oplus B}\ \oplus_{I1}$$

$$\frac{\Gamma \vdash M : B}{\Gamma \vdash \mathbf{inl}(M) : A \oplus B}\ \oplus_{I2}$$

$$\frac{\Delta \vdash M : A \oplus B \qquad \Gamma, x : A \vdash N_1 : C \qquad \Gamma, y : B \vdash N_2 : C}{\Gamma, \Delta \vdash \mathbf{case}\ M\ \mathbf{of}\ x \Rightarrow N_1\ ||y\ \Rightarrow N_2 : C}\ \oplus_E$$

Reduction rules:

- $\mathbf{case}\,\mathbf{inl}(M)\,\mathbf{of}\,x \Rightarrow N_1 \,||\, y \Rightarrow N_2 \to N_1[x := M]$,
- $\mathbf{case}\,\mathbf{inr}(M)\,\mathbf{of}\,x \Rightarrow N_1 \,||\, y \Rightarrow N_2 \to N_2[x := M]$,
- $\mathbf{case}\,M\,\mathbf{of}\,\mathbf{inl}(x) \Rightarrow \mathbf{inl}(x) \,||\, \mathbf{inr}(x) \Rightarrow \mathbf{inr}(x) \to M$,
- $N[z := \mathbf{case}\,M\,\mathbf{of}\,\mathbf{inl}(x) \Rightarrow N_1 \,||\, \mathbf{inr}(x) \Rightarrow N_2] \to \mathbf{case}\,M\,\mathbf{of}\,\mathbf{inl}(x) \Rightarrow N[z := N_1] \,||\, \mathbf{inr}(x) \Rightarrow N[z := N_2]$
-

4.3 Exponential Negation

Let $\neg A$ denote $A \multimap \bot$. The exponential negation is axiomatised with the following inference rules:

$$\dot{\neg}_I \frac{\begin{array}{c}\Gamma_1 \vdash M_1 : \dot{\neg}A_1 \\ \cdots \\ \Gamma_n \vdash M_n : \dot{\neg}A_n \qquad x_1 : \dot{\neg}A_1, \ldots, x_n : \dot{\neg}A_n \vdash N : \neg A\end{array}}{\Gamma_1, \ldots, \Gamma_n \vdash \dot{\neg}\, N\,\mathbf{with}\,\vec{x} = \vec{M} : \dot{\neg}A} \qquad \frac{\Gamma \vdash M : \dot{\neg}A}{\Gamma \vdash \mathbf{der}\,(M) : \neg A}\,\dot{\neg}_E$$

$$\frac{\Gamma \vdash M : \dot{\neg}A \qquad \Delta \vdash N : \bot}{\Gamma, \Delta \vdash \mathbf{del}(M; N) : \bot}\,\dot{\neg}_{\mathbf{W}} \qquad \frac{\Gamma \vdash M : \dot{\neg}A \qquad \Delta, x : \dot{\neg}A, y : \dot{\neg}A \vdash N : \bot}{\Gamma, \Delta \vdash \mathbf{let}\,x, y = M\,\mathbf{in}\,N : \bot}\,\dot{\neg}_{\mathbf{C}}$$

Reduction rules for $\dot{\neg}$: TODO: write down all those reduction rules tediously

5 Current Questions

- Do we need resource modalities? If so, then have a look at [MT10], [MTT18] and [MT07]. Exponential negation? Note sure what's more essential for the profinite setting.
- Should tensorial terms be symmetric (as well as dialogue categories)? At least, we'll have less boilerplate in the reduction rules.
- Do we need sums? Sums are essential for the resource-sensitive CPS as they model non-deterministic choice.
- When we model the profinite lambda terms, we use the category of finite sets as the starting point: intuitively, **FinSet** is the cartesian closed category of finite types. What's the analogue of **FinSet** for tensorial terms? $\mathbb{F}$-weighted automaton over Galois fields? Some references: [BR11], [MN23], [AMU18], [UACM16], [BBB$^+$12], [VZ14].
- How one can describe the constuction with the $\mathcal{P}$-functor from [JM18] for dialogue categories?

5.1 Parametricity for Tensorial Logic

The following subsection develops parametricity for the tensorial calculus in the fashion of [AMP25] and [ZZZ10].

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